Battle of the best

AI Fridays

PARTICIPANT ENGINEERING & PRODUCTION READINESS GUIDE

Build AI that is Ready for Enterprise

Mandatory engineering standard for every National Finale challenge — from business requirement to secure, traceable, resilient and deployable AI-native product.

Architecture · Security · Traceability · Testing · Exception Handling · Deployment

20
ENGINEERING GUIDE SECTIONS

23
DELIVERY ARTIFACTS

5
DELIVERY DAYS

100%
REQUIREMENT-TO-RUNTIME MINDSET

NAVIGATE

Engineering Guide Directory

Reimagine

Start with the industry outcome, user journey and a Human + AI + Software operating model.

Engineer

Build the complete application: experience, services, AI, data, integrations, security, controls and operations.

Prove

Demonstrate traceability, testing, failure recovery, security controls and a credible path to deployment.

SECTION 01

National Finale Mission & Standard

Build the first deployable version of an AI-powered enterprise product.

The National Finals are designed to test far more than the ability to build an AI prototype. Over five days, each team will take an industry problem, reimagine how it should work in an AI-native world, and engineer an end-to-end application with credible enterprise deployment readiness.

- National Finale Standard: Reimagine with AI. Engineer for Enterprise. Prove it in Production.

THE FINALE IS	THE FINALE IS NOT
Industry reimagination	A prompt-engineering competition
End-to-end product engineering	A chatbot-only hackathon
Production-minded AI delivery	A notebook or slide concept
Traceable engineering discipline	A happy-path scripted demo
Security, resilience and controls in the product	A governance checklist added at the end

SECTION 02

How to Reimagine the Industry Problem

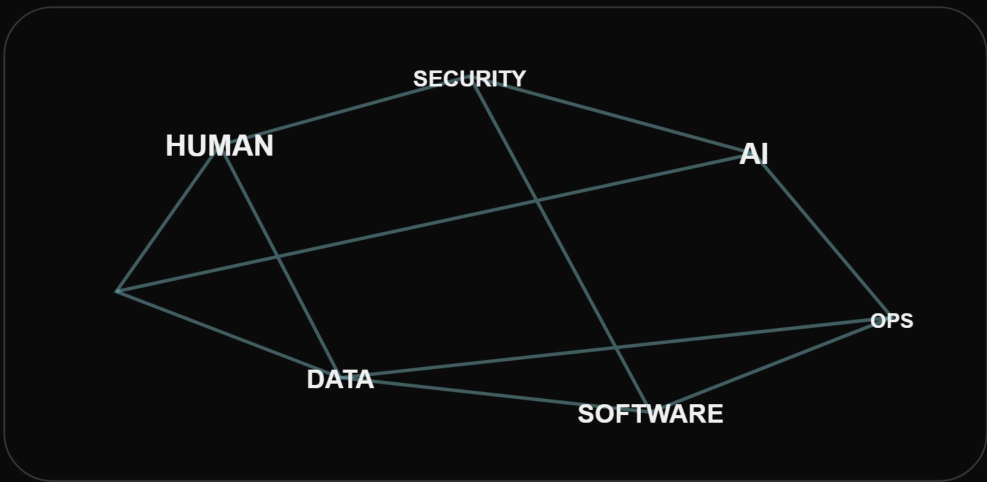
Do not begin with a preferred technology. Begin by asking what the industry process would look like if it were designed today around Humans + AI + Software.

- ✓ Challenge assumptions in the current process rather than automating every existing step.
- ✓ Remove unnecessary work and hand-offs where AI and software can change the operating model.
- ✓ Redesign the user experience so context, evidence and progress remain continuous across the journey.
- ✓ Identify capabilities that become possible only because of AI - understanding, reasoning, multimodal interpretation, orchestration or collaboration.
- ✓ Define the business outcome and acceptance criteria before selecting the technology pattern.

- Start with: What should become possible for the user and the enterprise?

SECTION 03

Human + AI + Software Design



The strongest solution will not necessarily automate everything. Teams must explicitly decide where humans lead, where AI augments, and where software executes predictable enterprise behaviour.

ROLE	PRIMARY RESPONSIBILITY
Humans	Judgement, accountability, empathy, exceptional decisions, approval and challenge.
AI	Understanding, reasoning, interpretation, generation, evidence synthesis, contextual assistance and orchestration.
Software	Predictable execution, integration, persistence, identity, transactions, workflow state, controls and scale.

Design one coherent operating system: Humans + AI + Software.

SECTION 04

Technology Freedom

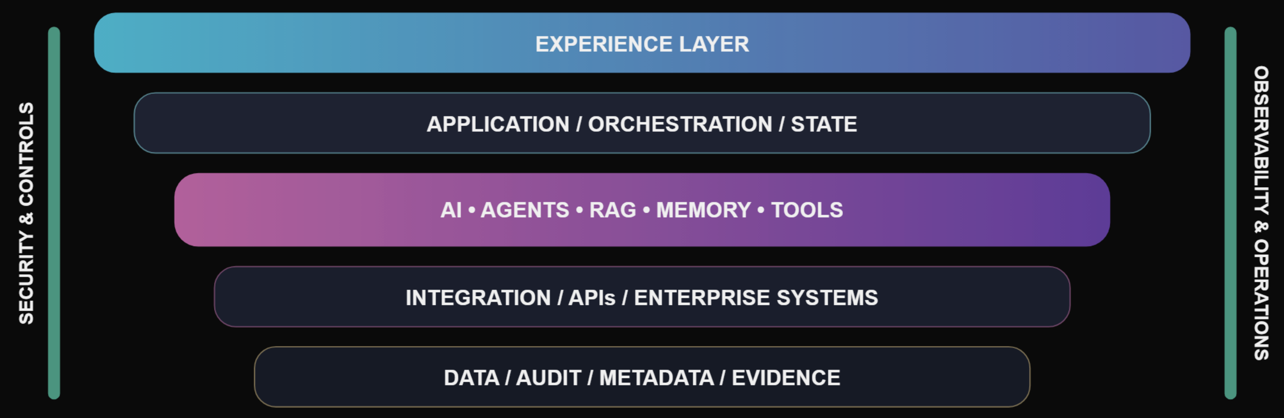
There is no mandated AI architecture or technology stack. Teams should select approaches that fit the problem and defend why they are appropriate.

- ✓ Large language and reasoning models
- ✓ Multimodal AI
- ✓ RAG and advanced retrieval
- ✓ Agentic and multi-agent systems
- ✓ Knowledge graphs and structured knowledge
- ✓ Memory and state architectures
- ✓ Tool calling and external APIs
- ✓ Workflow/orchestration engines
- ✓ Specification-driven development
- ✓ AI-assisted software engineering
- ✓ Other emerging AI techniques that materially improve the solution

Important: Technology novelty alone does not earn credit. The jury will evaluate appropriateness, implementation quality and business impact.

SECTION 05

Build the Complete Product



A notebook, prompt playground or chatbot connected to a vector store is not automatically an enterprise application. Your final implementation should demonstrate the meaningful layers required to operate the solution end-to-end.

LAYER	EXPECTED CAPABILITY
Experience layer	Web/mobile/conversational experience, case workspace or operational console.
Application layer	Business services, workflow/orchestration, case state and domain behaviour.
AI layer	Models, prompts/instructions, retrieval, agents, reasoning, context and memory.
Integration layer	Enterprise systems, APIs, tools, messages and external services.
Data layer	Application data, documents, persistent state, metadata and audit evidence.
Security & control layer	Identity, permissions, approvals, boundaries, policy enforcement and safeguards.
Operations layer	Logs, traces, metrics, evaluation, alerting, supportability and recovery evidence.

SECTION 06

Architecture & Engineering Standard

Your team must be able to explain and defend how the running product is structured, how responsibilities are separated and how the solution behaves under scale and failure.

- ✓ System boundaries, component responsibilities and service interactions.
- ✓ Data flows, model interactions, API/tool integration and event/message patterns.
- ✓ Persistent state, memory, long-running workflow behaviour and asynchronous processing.
- ✓ Identity propagation, trust boundaries and external dependencies.
- ✓ Scaling strategy, deployment topology, environment separation and configuration.
- ✓ Failure boundaries, retry/recovery patterns, idempotency where relevant and safe resumption.
- ✓ Observability sufficient to understand user, application, AI and tool behaviour.

Expect architecture defence questions: What happens at 50,000 users? What if a model provider fails? What if a workflow runs for days? What if an API succeeds but its response is lost? What if permissions change mid-flow?

SECTION 07

Security by Design

Security must exist in the implementation, not appear for the first time in the final presentation.

- ✓ Authentication, authorization, least privilege and role/attribute-based access as appropriate.
- ✓ Secrets management, encryption, session handling and secure API access.
- ✓ Sensitive-data classification, handling, retention and user/tenant isolation.
- ✓ Prompt injection, malicious document content, data exfiltration and inappropriate tool invocation.
- ✓ Dependency, package, container and API vulnerability considerations.
- ✓ Audit logging for material user, AI, tool and administrative actions.
- ✓ Clear trust boundaries between users, AI, enterprise services and external providers.

- Demonstrate controls: Where feasible, show the jury a control actively preventing an unauthorized or unsafe action.

SECTION 08

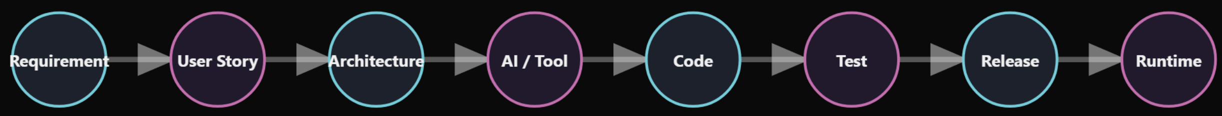
AI Controls & Human Accountability

Controls are product behaviour. Teams must explicitly define what AI can do, what requires human approval and what AI is prohibited from doing.

CONTROL CATEGORY	EXAMPLES
Allowed AI assistance	Retrieve, interpret, summarize, compare, generate, recommend or coordinate approved activities.
Human-approved actions	Consequential transactions, record changes, exceptional cases, sensitive communications or irreversible actions.
Prohibited AI actions	Unauthorized data access, bypassing approvals, unapproved tools, specified consequential decisions or unsafe actions.
Escalation	Insufficient evidence, low confidence, contradictory information, policy conflict, security concern or tool failure.
Auditability	Record AI recommendation, evidence used, tool/action requested, human approval/override and final outcome.

SECTION 09

Requirement-to-Production Traceability



Every change should remain traceable across the engineering lifecycle.

Traceability is a core National Finale expectation. The running product and the engineering artifacts must tell the same story.

- Trace the full chain: Business Problem -> Business Requirement -> User Story -> Acceptance Criteria -> Architecture Component -> AI Capability / Agent / Prompt / Tool -> Data / API -> Implementation -> Test Case -> Deployment Version -> Runtime Evidence.

The jury may select a requirement at random and ask where it originated, which component owns it, how it is implemented, what test proves it, what control protects it, and which deployed version contains it.

- ✓ Maintain traceability throughout development rather than creating an RTM manually at the end.
- ✓ When requirements change, identify impacted components, AI behaviours, tests and release artifacts.
- ✓ Preserve runtime evidence so the team can demonstrate that the deployed behaviour matches the requirement.

SECTION 10

SPECS / DSL - Specification-Driven AI Engineering

Teams are encouraged to experiment with structured specifications or a domain-specific language that connects business intent to AI/application behaviour and tests.

- Example relationship: REQ-017 -> Investigator -> Evidence Comparison -> Investigation Agent -> Document API -> Case Evidence -> READ_ONLY -> HUMAN_APPROVAL -> TOOL_FAILURE_ESCALATE -> AC-017.2 -> TC-017.2

A strong SPECS / DSL approach may help generate or validate parts of the implementation, agent configuration, workflow, APIs, test cases, documentation, controls and traceability. The format is not prescribed; innovation is encouraged.

SECTION 11

Exception Handling & Resilience

Enterprise applications cannot assume the happy path. Failure behaviour should be intentionally designed, implemented and tested.

- ✓ Missing, malformed, contradictory, duplicated or outdated information.
- ✓ Tool or API unavailability, timeout, stale response or uncertain external state.
- ✓ Authentication/authorization failure and changed permissions.
- ✓ Model uncertainty, refusal, failure or unsupported conclusion.
- ✓ Human rejection, override or change of objective.
- ✓ Partial completion, duplicate action, interrupted workflow and retry/recovery.
- ✓ Superseded documents, requirement changes and security incidents.

- Design principle: A failure should lead to a safe, observable and recoverable state - not an unexplained dead end.

SECTION 12

State, Memory & Resumability

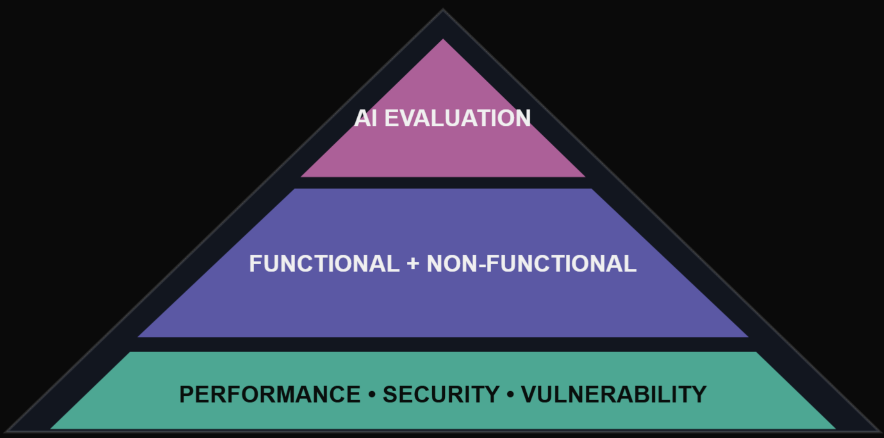
Conversation history is not the same as enterprise state. Important business state must survive beyond a single model turn or session.

- ✓ Persistent case/workflow state
- ✓ User and role context
- ✓ Versioned evidence and source lineage
- ✓ Agent/tool actions and outcomes
- ✓ Human approvals, overrides and comments
- ✓ Checkpoints and resumability
- ✓ Audit history
- ✓ Model/prompt/configuration version where materially relevant

- Jury challenge: A workflow may be interrupted and resumed by another user. Your application should continue safely with the correct state and permissions.

SECTION 13

Testing & Evaluation



The National Finals expect evidence that the product works, behaves safely and can withstand unseen scenarios.

TEST AREA	MINIMUM EXPECTATION
Functional	Positive, negative, alternate flow, boundary and exception scenarios mapped to requirements.
AI evaluation	Correctness, grounding, provenance, hallucination, tool selection, escalation, agent behaviour and task completion as applicable.
Non-functional	Reliability, availability, scalability, usability, accessibility, maintainability, recoverability and observability.
Performance	Latency, throughput, concurrency, long-running workflows and model/API behaviour under realistic or simulated load.
Security	Authentication, authorization, isolation, malicious input, prompt injection, tool misuse, data leakage and secrets handling.
Vulnerability	Assessment of dependencies, packages, APIs, application/container components with severity and remediation status.

Create a reusable test set and test execution report. Expect the jury to introduce unseen test cases and ask you to show which requirement each test validates.

SECTION 14

Observability & Operations

A production-ready AI application must allow operators and engineers to understand what happened and why.

- ✓ Application logs and business-event logs.
- ✓ Agent/model/tool execution traces where appropriate.
- ✓ User actions, approvals and overrides.
- ✓ Errors, retries, failures and recovery actions.
- ✓ Latency, health indicators and important operational metrics.
- ✓ Audit evidence for consequential actions.
- ✓ Troubleshooting paths and operational recovery procedures.

SECTION 15 · PART 1 OF 3

Mandatory Delivery Artifacts

Artifacts may be combined, but the information must exist, remain consistent and be inspectable.

Teams may use AI extensively to create, maintain and validate these artifacts. The team remains accountable for accuracy, completeness and consistency with the running application.

#	ARTIFACT	MINIMUM EXPECTATION
01	**Business Requirements Document (BRD)**	Business context, objectives, stakeholders/personas, current-state pain points, future-state vision, scope, assumptions, constraints, requirements, success measures and key rules.
02	**AI-Native Future-State Process / Journey**	How the process is reimagined using Humans + AI + Software; what is removed, redesigned, augmented or automated.
03	**User Stories & Acceptance Criteria**	Prioritized epics/features/stories with measurable functional, AI-specific, negative and exception acceptance criteria.
04	**Functional / System Specification**	Application behaviour, workflows, states, validations, alternate paths, error handling and external interactions.
05	**SPECS / DSL**	Structured or machine-readable specification covering AI capabilities, agents, tools, permissions, inputs/outputs, state, controls, exceptions and tests.
06	**Solution Architecture**	End-to-end logical architecture across experience, application, AI, integration, data, security/control and operations layers.
07	**Technical / Component Architecture**	Major services/components, responsibilities, APIs/events, state ownership, interaction patterns and failure boundaries.
08	**AI / Agent Architecture**	Models, agents, prompts/instructions, retrieval, memory/context, orchestration, tool calls, human-in-the-loop and safeguards.
09	**Deployment Architecture**	Target topology, environments, gateways, networking, storage, model endpoints, identity, secrets, monitoring and scaling approach.
10	**Database & Data Design**	Conceptual/logical model, entities, relationships, schema, application/agent state, audit data, metadata and retention assumptions.

SECTION 15 · PART 2 OF 3

Mandatory Delivery Artifacts — Continued

Artifacts may be combined, but the information must exist, remain consistent and be inspectable.

Continued delivery artifacts — items 11 to 19.

#	ARTIFACT	MINIMUM EXPECTATION
11	**API / Integration Specifications**	Contracts, request/response/event structures, authentication/authorization, timeout, retry and failure behaviour.
12	**Security Architecture & Threat Model**	Trust boundaries, identities/roles, sensitive data, attack surfaces, AI-specific threats, mitigations and residual risks.
13	**AI Controls & Human Decision Matrix**	What AI may retrieve/recommend/execute, what needs human approval, what is prohibited, confidence/evidence boundaries and escalation.
14	**Functional Test Cases**	Positive, negative, alternate-flow, boundary and exception tests mapped to requirements and acceptance criteria.
15	**AI Evaluation Test Suite**	Grounding, correctness, hallucination, source attribution, tool selection, agent behaviour, escalation and task completion tests as applicable.
16	**Non-Functional Test Cases**	Reliability, availability, scalability, usability, accessibility, maintainability, recoverability and observability tests.
17	**Performance Test Cases & Results**	Latency, throughput, concurrency, long-running workflow and model/API performance under realistic or simulated load.
18	**Security Test Cases**	Authentication, authorization, isolation, malicious input, prompt injection, inappropriate tool invocation, data leakage and secrets exposure.
19	**Vulnerability Assessment**	Application/API/dependency/container assessment with severity, impact, mitigation and status for material findings.

SECTION 15 · PART 3 OF 3

Mandatory Delivery Artifacts — Continued

Artifacts may be combined, but the information must exist, remain consistent and be inspectable.

Continued delivery artifacts — items 20 to 23.

#	ARTIFACT	MINIMUM EXPECTATION
20	**Requirements Traceability Matrix (RTM)**	Trace business requirements through stories, acceptance criteria, components, AI capabilities, data/APIs, tests, release and runtime evidence.
21	**Test Execution Report**	Tests planned/executed, pass/fail, defects, fixes, retest evidence, deferred items and known limitations.
22	**Deployment / Release Package**	Deployable application/package, configuration, environment requirements, version information, dependencies and rollback/recovery approach.
23	**User Guide**	Core user journeys, access, approvals, exceptions, warnings, help paths and known limitations.

Artifact Quality Rules

- ✓ Do not create disconnected documents merely to satisfy a checklist. Combine artifacts where it improves coherence.
- ✓ Artifacts should evolve with the product. A changed requirement should change the relevant story, specification, implementation, test and traceability evidence.
- ✓ Every important design claim should be demonstrable in the running solution or clearly marked as a production-hardening item.
- ✓ Known limitations and technical debt should be explicit. Production readiness does not mean claiming perfection.

The Traceability Challenge: The jury may ask you to choose any requirement and show its path from BRD to user story, component, AI behaviour, data/API, test, release and runtime evidence.

SECTION 16

Final Demonstration Standard

- ✓ Business problem: why the industry problem matters and what is broken today.
- ✓ Reimagined experience: what has fundamentally changed in the AI-native future state.
- ✓ AI innovation: what becomes possible through AI that was previously impractical.
- ✓ End-to-end product: run the meaningful journey through real application components.
- ✓ Human control: show where accountability, approval and challenge remain.
- ✓ Architecture: explain how the product works, scales and fails safely.
- ✓ Security: demonstrate at least one meaningful security/control boundary.
- ✓ Traceability: pick a requirement and follow it to runtime evidence.
- ✓ Resilience: demonstrate a non-happy-path scenario and recovery.
- ✓ Testing: show executed test evidence, including AI, non-functional and security coverage.
- ✓ Deployment: explain how the application is packaged, configured and released.
- ✓ Business impact: show the expected industry outcome and credible adoption path.

SECTION 17

Evaluation Framework

EVALUATION AREA	WEIGHT	WHAT EXCELLENT LOOKS LIKE
Problem Reimagination & Industry Impact	15%	Deep problem understanding and a meaningfully redesigned future state.
Innovation & AI-Native Thinking	15%	Original, appropriate AI use that goes beyond obvious automation.
End-to-End Working Product	20%	Usable, integrated and demonstrably working across the core journey.
Architecture & Engineering Quality	15%	Coherent, maintainable, scalable and deployment-credible engineering.
Security, Responsible AI & Controls	10%	Controls exist in the product and protect consequential boundaries.
Requirement-to-Production Traceability	10%	Consistent artifacts and demonstrable requirement-to-runtime evidence.
Exception Handling & Resilience	10%	Safe behaviour, recovery, adaptation and resumability under failure/change.
Executive Articulation	5%	Clear value story, architecture defence and credible path to production.

- Winning balance: A beautiful AI demo with weak engineering cannot win. A conventional application with superficial AI cannot win. The winning solution combines innovation, AI, software engineering, controls, traceability and industry impact.

SECTION 18

Final Submission Checklist

☐ RUNNING END-TO-END APPLICATION	☐ SOURCE CODE / SOLUTION PACKAGE
☐ Business Requirements Document (BRD)	☐ AI-native future-state journey
☐ User stories and acceptance criteria	☐ Functional/System Specification
☐ SPECS / DSL	☐ Solution Architecture
☐ Technical / Component Architecture	☐ AI / Agent Architecture
☐ Deployment Architecture	☐ Database / Data Design
☐ API / Integration Specifications	☐ Security Architecture and Threat Model
☐ AI Control / Human Decision Matrix	☐ Functional Test Cases
☐ AI Evaluation Test Suite	☐ Non-Functional Test Cases
☐ Performance Test Cases & Results	☐ Security Test Cases
☐ Vulnerability Assessment	☐ Test Execution Report
☐ Requirements Traceability Matrix	☐ Deployment / Release Package
☐ User Guide	

- Final challenge: Do not show us what AI can generate. Show us what becomes possible when Humans, AI and Software are engineered to work together.

AI Friday National Finals

Reimagine with AI. Engineer for Enterprise. Prove it in Production.